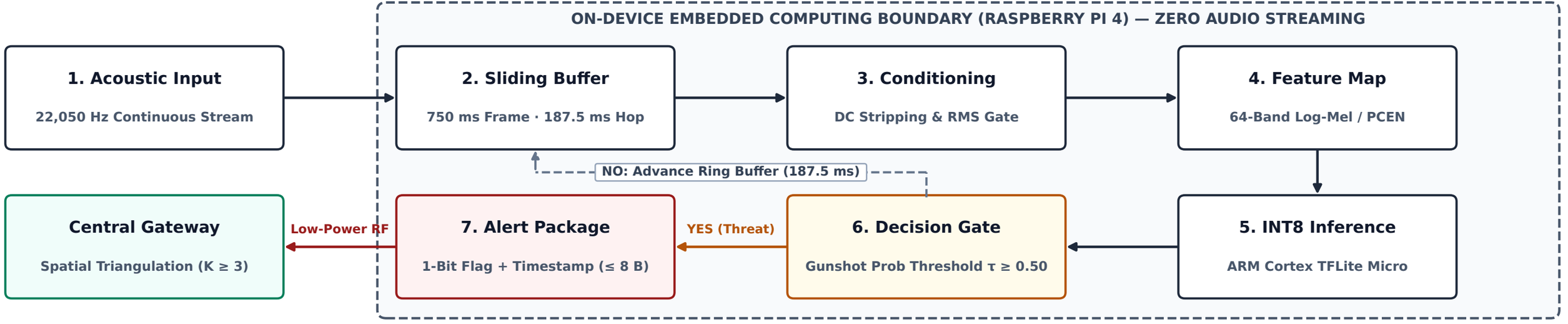
ON-DEVICE EMBEDDED COMPUTING BOUNDARY (RASPBERRY PI 4) — ZERO AUDIO STREAMING
1. Acoustic Input
22,050 Hz Continuous Stream
2. Sliding Buffer
750 ms Frame · 187.5 ms Hop
3. Conditioning
DC Stripping & RMS Gate
4. Feature Map
64-Band Log-Mel / PCEN
5. INT8 Inference
ARM Cortex TFLite Micro
6. Decision Gate
Gunshot Prob Threshold τ ≥ 0.50
NO: Advance Ring Buffer (187.5 ms)
YES (Threat)
7. Alert Package
1-Bit Flag + Timestamp (≤ 8 B)
Low-Power RF
Central Gateway
Spatial Triangulation (K ≥ 3)